D1 — Proposal and Feasibility

Project: BloodLink PK — Blood Donor and Donation Camp Coordination System **Course:** Software Engineering (CT-365), Complex Computing Problem **Team:** Group [XX] — [Member 1, Roll No.] · [Member 2, Roll No.] **Repository:** [Git URL]

1. Problem Statement

Pakistan needs over 5 million blood donations a year but receives only about 2.7 million, and WHO reports that only 18% are voluntary, with 82% coming from family or replacement donors. In practice, a patient's family must find a donor themselves, usually through unstructured WhatsApp and Facebook posts that cannot be verified, are never marked as fulfilled, and reach donors without any eligibility check. Donors are rarely tracked or re-engaged after donating, and camp organizers coordinate on paper or in chat groups.

Existing apps (e.g., Madadgaar, Bloodou, and several university prototypes) focus on matching donors to requests. They do not cover **verified request lifecycles, eligibility-aware donor retention, or end-to-end camp management**. This project addresses that gap.

2. Objectives

1. **Faster matching:** make blood requests structured, time-bound and verifiable, and match them to compatible, eligible, nearby donors.
2. **Donor retention:** track donation history and eligibility, and remind donors when they can donate again.
3. **Camp coordination:** let organizers create camps, manage slot booking and capacity, and record attendance and donations.

3. Stakeholders

Stakeholder	Interest
Donors	Easy registration, relevant requests, knowing when they can donate

Stakeholder	Interest
Patients / families (requesters)	Quickly reaching eligible donors
Hospitals / blood banks	Verified requests, organized donor supply
Camp organizers (NGOs, university societies)	Slot booking, attendance and donation records
System administrator	Moderation, user management, reports
Instructor / evaluators	Evidence of sound engineering process

4. Scope

In scope

- Four roles with role-based access: Donor, Requester, Organizer/Blood bank staff, Admin
- Registration, login, and donor profile (blood group, city, last donation date)
- Blood requests with status (open, fulfilled, expired) and expiry time
- Rule-based matching on blood-group compatibility, city and eligibility
- Camp creation, slot booking with capacity limits, attendance and donation recording
- Eligibility calculation and "eligible again" reminders
- Admin dashboard and request moderation
- One-tap WhatsApp share link for a request

Out of scope: payments, native mobile app, live GPS tracking, SMS gateway, medical screening or diagnosis, hospital-system integration, blood inventory management.

Assumptions and constraints: 2–3 week timeline, two-person team, free-tier tools only, test data used for the prototype, eligibility rules to be confirmed against an official guideline and kept configurable.

5. Feasibility

Type	Assessment
Technical	Feasible: standard web stack with a relational database, matching the team's skills
Schedule	Feasible: focused core fits two sprints plus testing
Economic	Feasible: free hosting and open-source tools, no paid services
Operational	Feasible: pilot with seeded data and a few volunteer users

6. Risks

Risk	Mitigation
Scope creep	Freeze core scope in week 1; keep extras in a backlog
Donor privacy (phone numbers, health data)	Collect minimum data, hide contact details until the donor accepts, enforce role-based access
Fake or abusive requests	Expiry on every request, rate limits, admin moderation
Incorrect eligibility rules	Verify against an official guideline; keep rules configurable
Small team, short timeline	Weekly milestones, module ownership, peer code review
No real users for testing	Seed realistic data; small pilot

7. SDLC Model: Iterative–Incremental with Scrum-style Sprints

Justification: requirements may shift as the gap is validated, which rules out Waterfall. D4 requires at least two iterations, and this model delivers working increments each sprint. Full Scrum is too heavy for two people, so we use a short sprint plan, a shared task board,

and a review at the end of each sprint.

Increment	Deliverable
Sprint 1	Authentication and roles, donor profiles, blood requests, matching
Sprint 2	Camps and slot booking, eligibility reminders, admin dashboard
Final week	Testing, defect fixes, documentation, demo

Sources for statistics: WHO Pakistan statements on World Blood Donor Day (via APP); LUMS Madadgaar project page.